



My RSE Journey

I kind of did it my way ...

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1. Where do I come from?
2. How did I get here?
 - ▶ Phase 1: South Africa
 - ▶ Phase 2: United Kingdom
 - ▶ Phase 3: Back to university
 - ▶ Phase 4: Post PhD





Potchefstroom University vir CHO

It is now called

North West University



BA with Tswana, Sociology, Zulu,
Anthropology
Post graduate Higher Education Diploma



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- ▶ Logic Systems (Digital Electronics)
- ▶ Basic Programming
- ▶ Maths
- ▶ Science
- ▶ Computer Practise and typing technique



- ▶ Programmer at a Municipal Pension Fund using Basic
- ▶ Programmer at Securities Trading and Technologies using UCSD Pascal
- ▶ Took two years out
 - ▶ Picked oysters in Knysna down in the Cape
 - ▶ Worked as a fireman and paramedic
 - ▶ Was a Red Cross First Aid instructor
 - ▶ Wrote text books for Computer Practice and Typing Technique with my mom because there weren't any



- ▶ Programmer at a Municipal Pension Fund writing portfolio management software using Basic
- ▶ Programmer at Securities Trading and Technologies (STT) using UCSD Pascal
- ▶ Took two years out
 - ▶ Picked oysters in Knysna down in the Cape
 - ▶ Worked as a fireman and paramedic
 - ▶ Was a Red Cross First Aid instructor
 - ▶ Wrote text books for Computer Practice and Typing Technique with my mom because there weren't any
- ▶ Savings ran out - back to STT



- ▶ I asked for more money after a couple of years and the boss shipped me off to SAFEX (the South African Futures Exchange) for two years as IT Manager - can't say that I like it much.
 - ▶ Finished the Diploma in Datametrics
 - ▶ Became a certified Netware Engineer (Novell) - because the work paid!
 - ▶ Posted my CV on the Internet



04:34:17 (CET)
07, November 1996

Snowball
SFr 993.00

Spot the Ball
SFr 477.00

Little BIG
SFr 248.00

LOTTO SHOOT OUT
SFr 27.00

Weekly Lotto (Game 54) takes place on 8 November 1996

Winning Numbers (Game 53)
7, 15, 16, 18, 21, 26

Microsoft Internet Explorer

You are connected **INSECURELY** to InterLotto
(Switch to Secure)



Authorised and controlled by the Liechtenstein Government



[Lottery Games](#) | [Your Account](#) | [Information](#) | [Instant Games](#) | [New Users](#) | [Help](#) | [EGG](#)

Ittai B__ just won SFr 2999 in Snowball!

Guaranteed winner? - try Lotto Shoot Out (one game every six hours)

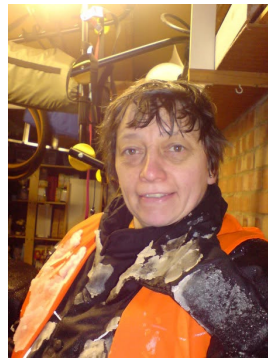
Easier odds? play Little Big One - Lottery Games

[English](#) | [French](#) | [German](#) | [Spanish](#) | [Italian](#) | [Chinese](#)



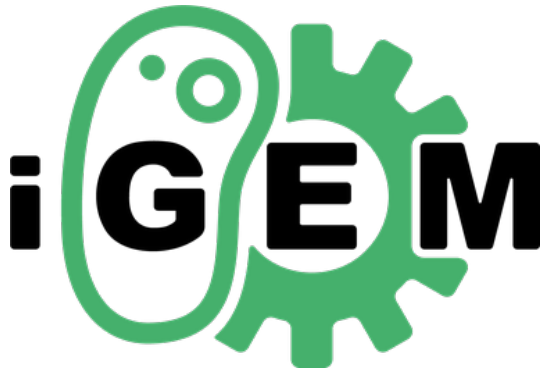
Over the next few years ...

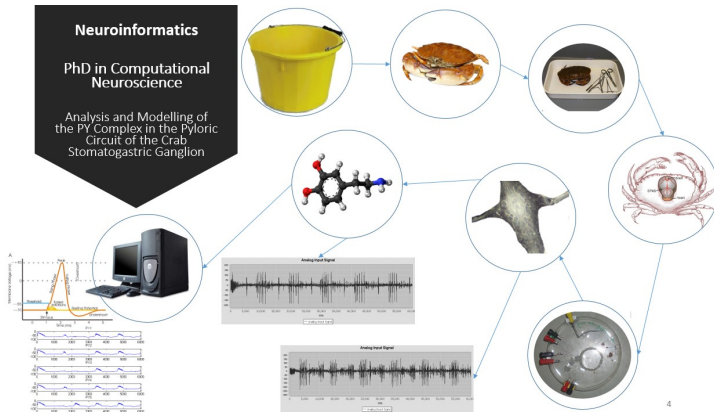
- ▶ Got married
- ▶ Moved to Durham
- ▶ Had two kids
- ▶ Got divorced
- ▶ Worked at a college in Durham for a couple of years supporting Blackboard and developing their websites
- ▶ Worked at Serco for the next five years - didn't like that
- ▶ Decided to go back to uni full time





- ▶ 2009-10 MSc Bioinformatics and Computational Systems Biology
- ▶ Took part in iGEM 2010
- ▶ Hard work, but it was fun and I enjoyed it







- ▶ Two years at the Bioinformatics Support Unit at Newcastle
- ▶ Senior Research Associate in Computer Science
- ▶ 2019 - the RSE team is established and I apply
- ▶ Each RSE has at least two projects



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- ▶ 2020 certified Carpentries Instructor
- ▶ 2021 certified Instructor Trainer
- ▶ Start spending half my time on Carpentries training
- ▶ 2022 I get an SSI fellowship for CarpentriesOffline





- ▶ Our team has grown from two (pre-2019) to 27 (2026)
- ▶ 2024 - Planning starts for a new HPC
- ▶ 2025 - Installation of Comet starts
- ▶ I go full-time on training and community and I get a minion to help me
- ▶ 2026 - Comet eventually up an running



Carpentries Workshops

- ▶ 500 workshop attendees
- ▶ 39 separate events
- ▶ 2-hour online HPC inductions to 2-day in-person 'Better Research Software' workshops
- ▶ Instructors and helpers: 25 different people from the RSE team and beyond



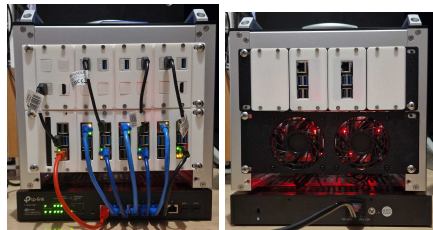
CarpentriesOffline

- ▶ SSI Collaborations Workshop 2021
- ▶ Running workshops without Internet access
- ▶ Use Raspberry Pi as an access point and web server
- ▶ Hackday winner
- ▶ SSI Fellowship 2022





- ▶ Pixie: Raspberry Pi 4s, one login node, five compute nodes, one access point
- ▶ The same open source software stack as many HPCs: Slurm, lmod, munge
- ▶ Running Workshops and Hackerthons
 - ▶ Lesson to build a login node and a compute node
 - ▶ Adapt HPC lesson for Pixie
 - ▶ Networking lesson



Front and Back





Me on a page outside of work



Resources



- ▶ <https://www.flaticon.com/free-icons/crab>
- ▶ <https://www.flaticon.com/free-icons/firefighter>
- ▶ <https://www.vecteezy.com/free-vector/teacher-icon>
- ▶ <https://thenounproject.com/browse/icons/term/children/>
- ▶ <https://thenounproject.com/browse/icons/term/lottery-tickets/>
- ▶ <https://thenounproject.com/browse/icons/term/programming/>
- ▶ PUK Hoofgebou: <https://www.flickr.com/photos/riaan/>
- ▶ Oysters: <https://www.knysnaoysterfestival.co.za/>
- ▶ Interlotto: <https://www.computerhistory.org/revolution/artifact/392/2344>